

Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that

1. Accept a Linux command as input.
2. Create a child process using `fork()`.
3. Execute the command in the child process using an appropriate `exec()` system call.
4. Allow the parent process to wait for the child using `wait ()`.
5. Display the Process ID (PID) of both parent and child processes.

Code:

```
geya@LenovoL480: ~/OS/Pra x + v
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    char command[50];
    pid_t pid;

    printf("Enter Linux command: ");
    scanf("%49s", command);

    pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
    }
    else if (pid == 0) {
        // Child process
        printf("Child PID: %d\n", getpid());

        execlp(command, command, NULL);

        printf("Command execution failed!\n");
    }
    else {
        // Parent process
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

Output:

```
geya@LenovoL480: ~/OS/Pra x + v
geya@LenovoL480:~/OS/Practical/Week 1$ gcc Task_1.c -o Task_1
geya@LenovoL480:~/OS/Practical/Week 1$ ./Task_1
Enter Linux command: ls
Parent PID: 671
Child PID: 672
Child PID: 672
Task_1 Task_1.c
Child process completed.
geya@LenovoL480:~/OS/Practical/Week 1$ |
```